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The fastest repetition in a set predicts the number of repetitions completed to failure during resistance training: The impact of individual characteristics

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ABSTRACT

The aim of this study was to examine the goodness of fit and prediction accuracy of general and individual relationships between the maximum number of repetitions completed in a set (XRM) and the fastest repetition in the set (XRM-velocity relationship) in the free-weight back squat exercise. The effects of sex, training status and history, as well as personality traits, on the goodness of fit and the accuracy of these relationships were also investigated. Forty-six resistance-trained people (15 females and 31 males) performed a one-repetition maximum (1RM) test, and two repetitions to failure (RTF) tests, 72 h apart. RTF tests were performed with 70, 80, and 90% of 1RM with 10 min of inter-set rest. A greater goodness of fit was found for individual XRM-velocity relationships (R^2 = 0.98; residual standard error [RSE] = 1.01–1.15) compared to general XRM-velocity relationships (R^2 = 0.45–0.49; RSE = 3.44–3.66). Individual, but not general XRM-velocity relationships established in the first testing session yielded acceptable prediction accuracy (mean error = 1.96 vs 2.81 repetitions) of XRM in the subsequent testing session. Sex, training status and history, and personality traits did not affect the goodness of fit of general and individual XRM-velocity relationships or their prediction accuracy, suggesting the potential generalisability of those findings among resistance-trained populations. Therefore, individual XRM-velocity relationships can be used to prescribe loads matching an intended XRM by recording only the fastest velocity of the set with a given load and predicting, with reasonable accuracy, the XRM for the individual.

1. Introduction

Resistance training is performed by athletes to optimise strength, power and hypertrophy [1]. Additionally, it is recommended by national health organisations for improving health in the wider population [2,3]. Designing efficient, individualised RT programs requires considering many variables such as exercise selection, order, intensity, volume, and others. However, intensity and volume are considered critical in program design due to their profound effects on the magnitude of RT-induced adaptations [4,5].

Traditionally, RT intensity is prescribed via percentage of one-repetition maximum (1RM; e.g., 85% of 1RM) or a load that allows a given number of repetitions before reaching muscle failure (XRM; e.g., 12RM is the load that can be lifted exactly 12 times). Unfortunately, establishing 1RM or XRM for the purposes of RT monitoring and prescription can be physically and psychologically demanding and time consuming (e.g., compromised safety, frequent testing for different exercises) [6]. Moreover, if 1RM and XRM are not re-evaluated frequently, they become outdated as training adaptations occur. Both RT prescription methods can also under- or over-estimate individuals' abilities on a given day, even when re-evaluated periodically, due to normal fluctuations in readiness to train caused by environmental, lifestyle, and other training factors. Thus, researchers search for alternatives to 1RM and XRM prescription methods.

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One of the proposed alternatives for prescribing RT intensity consists of recording barbell velocity during incremental loading tests to establish a load-velocity profile [7,8]. Indeed, many researchers report an inverse, linear relationship between barbell velocity and%1RM [7,9], although a minority of their studies use free-weight exercises [8,10]. One rationale for monitoring repetition velocity relies upon its stability at a given load [10], meaning any fluctuation in velocity beyond the normal variation observed between training sessions likely reflects fatigue or strength gains [10]. Thus, monitoring velocity may help individualise training prescription. Further, researchers suggest individual- rather than general load-velocity profiles be established because %1RM-velocity relationships are highly individual [8,11,12].

Despite the benefits of establishing an individual load-velocity profile for monitoring and prescribing RT, some professionals still prefer the use of the XRM-based method as it guarantees the knowledge of the number of repetitions left in reserve by individuals with a given load. Additionally, compared to other RT prescription methods, knowing an individual's XRM provides insight into the training volume completed. While there are different ways of monitoring volume, it is usually defined as the number of sets and repetitions in a training session. Thus, if sets are fixed, volume is largely determined by repetitions per set (i.e., set volume). Set volume can be monitored by prescribing the XRM of a given load for an individual. However, it must be noted that researchers previously reported [13] the number of repetitions completed to failure (RTF) is highly individual depending on training background and the %1RM used; though individual load-velocity relationships may reduce this variability [14,15]. Despite efforts to establish %1RM-velocity relationships with different exercises, little attention has been given to the potential XRM-velocity relationship. This relationship, if proven valid, could allow for daily XRM prediction – and therefore easier and more accurate RT prescription and monitoring of the training volume completed – without frequent RTF testing.

Only two previous studies [16,17] attempted to establish XRM-velocity relationships using maximal velocity of each set to failure to predict different XRMs. The authors reported that individual, rather than general (i.e., pooling the data from all individuals) XRM-velocity relationships provided a better goodness of fit for both Smith machine bench press and bench-pull exercises, allowing better XRM prediction accuracy [16,17]. However, the XRM-velocity relationship for free-weight exercises such as the back squat is unexplored, which is important given the popularity of free-weight exercises among trainees, especially athletes. Other methodological factors should also be considered with the XRM-velocity relationship. Notably, the predictive validity of such relationships has not been examined in free-weight exercises with multiple loads. Unless an initially established XRM-velocity relationship can predict a subsequent XRM with acceptable accuracy, it suffers the same drawbacks as traditional, non-velocity-based RT prescriptions (e.g., normal daily fluctuations affect the relationship's accuracy). Moreover, it is unknown whether sex, training status and history affect the XRM-velocity relationship's stability. Personality traits such as emotional stability and conscientiousness can influence how individuals cope with fatigue [18], thus potentially affecting the XRM-velocity relationship's accuracy. Therefore, many factors related to the XRM-velocity relationship are yet unexplored.

To address the above-mentioned scarcity in the literature, this study (1) examined the goodness of fit of general and individual XRM-velocity relationships during free-weight back squats; (2) determined the effects of sex, training status and history, and personality traits on the models' fit; and (3) determined whether general and individual XRM-velocity relationships accurately predicted subsequent testing session data and examined factors potentially affecting this accuracy. Based on previous studies [16,17], we hypothesised a greater goodness of fit for individual XRM-velocity relationships, whereas no other hypotheses were formulated given the dearth of literature on the topic.

2. Materials and methods

2.1. Design

A repeated-measures, crossover design was used to investigate the predictive ability of the maximal velocities of three sets to failure with different loads in the free-weight back squat exercise for determining XRM. Before RTF sessions, two preliminary sessions occurred including a familiarisation session with all procedures and conditions, and a 1RM free-weight back squat assessment. Thereafter, participants reported to the laboratory on two separate occasions and performed a single set with 90, 80, and 70% of their pre-determined 1RM with 10 min rest between sets, and 72 h rest between sessions. To control for environmental and diurnal factors, the same researchers were always present, and all sessions occurred at the same time (± 1 h) under similar (~20 °C and ~60% humidity) conditions.

2.2. Participants

Fifty-one resistance-trained people (15 females and 36 males; 18–40 years old) were initially recruited. Unfortunately, five males withdrew due to injury outside the study (n = 3) or for undisclosed personal reasons (n = 2). These participants' data were not analyzed as the purpose of this study was to determine the prediction accuracy of the XRM-velocity relationship. 1RM relative to body mass ranged from 0.86–2.12 to 1.18–2.61 for females and males, respectively. More participant data are reported in Supplementary file I. Participants had to confirm they (1) were willing to cease additional lower-body training during the study; (2) were not taking any medications related to metabolic or cardiovascular functioning; (3) were not injured; (4) were not taking anabolic steroids and did not have a history of their use; and (5) had a least 6 months of RT experience, training at least twice per week, performing back squats at least once a week, without more than two weeks off during that period. Before commencing, participants gave informed, written consent. The University Ethics Committee approved the study protocol (approval number: 20/55).

2.3. Familiarisation session

Upon arrival, participants completed a training history questionnaire, then provided training logs for their most recent, heaviest back squat session, and conservatively estimated their 1RM to inform warm-up loads for the subsequent 1RM session. Their body mass and height were then recorded using an electronic scale and wall mounted stadiometer (Seca Ltd, Hamburg, Germany). Thereafter, participants completed a standardised warm-up of cycling at 100 rpm for 5 min and dynamic stretching for 2 min; 10 bodyweight lunges and squats; and 10 barbell-only squats. Participants were then familiarised with the visual feedback on a screen showing mean barbell velocity and instructions (1) to descend into a squat at a controlled velocity, to then lift the barbell up as fast as they can, and (2) to take at least a momentary pause between repetitions, not lasting more than 2 s, while remaining in contact with the floor (i.e., no jumping or lifting the heels). To ensure familiarity with the instructions and visual feedback, participants then completed 3 repetitions at 20, 40, and 60%, then 10 repetitions at 60% of estimated 1RM under those conditions. All participants performed at least two sets with consistent repetition velocities (± 0.02 m/s) and reported they felt comfortable with the instructions and visual feedback.

2.4. One repetition maximum session

Participants performed the same standardised warm-up before the 1RM assessment. Thereafter, participants performed 3 repetitions at 20, 40, and 60%; and 1 repetition at 80 and 90% of estimated 1RM, followed by 1RM attempts using a 20 kg barbell (Rogue, Columbus, Ohio, USA) and calibrated weight plates (Eleiko, Halmstad, Sweden, EU) [8,10,19,

20]. After successful 1RM attempts, load increased 1–12.5 kg in consultation with the participants. 1RM tests terminated when participants could not complete an attempt or when a completed attempt required compromised movement technique. A maximum of five 1RM attempts were allowed. Three- and four-minutes rest were provided between submaximal sets and 1RM attempts, respectively. Participants adopted self-selected foot stances and eccentric tempos. They received the same lifting instructions as during the familiarisation session and were verbally encouraged and had a screen in front of them providing visual barbell velocity feedback. For a repetition to be considered successful, the top of the participants' thighs had to be at least parallel to the floor, which was visually confirmed by the researchers, and a camera perpendicular to the participant.

2.5. Repetitions to failure sessions

Before RTF tests, participants performed the same standardised warm-up as during previous sessions. Thereafter, participants completed 10, 5, 3, and 1 repetitions of the free-weight back squat against 30, 50, 70, and 90% of their established 90% of 1RM (i.e., the heaviest load for the day), respectively. Three minutes rest between warm-up sets and 4 min prior to the first set to failure was provided. Participants performed RTF from the highest to lowest load (i.e., 90% first, 80% second, and 70% 1RM third) in a fixed sequence to prevent excessive fatigue accumulation – caused by the high-repetition test with 70% – from negatively affecting RTF with 80 and 90%. To ensure recovery, 10 min rest was provided between sets and 72 h between sessions. The same lifting instructions, verbal encouragement, and visual feedback standardisation previously described also occurred during RTF sessions.

2.6. Data acquisition

To examine habitual RT practices of the participants, they were asked about (1) the number of repetitions they perform in training, on average; (2) the intensity of load at which they usually train; (3) the usual number of repetitions they leave in reserve at the end of sets, and (4) their training experience. The full training history questionnaire is presented in supplementary file II. Questions were multi-choice, and thus responses were treated as categorical. Response frequency was inspected for each category (overall and within levels of outcome variables used in models) and these variables were recorded by merging several response options to avoid having less than 5 responses in more than 20% of cells [21]. Specifically, the number of repetitions performed (1–5, 5–8, and >8) and load (<70, 70–80, 80–90) were transformed into categorical variables containing three levels. The number of repetitions in reserve (0–2 and 2–4) and RT experience (≤3 and >3) were transformed into categorical variables containing two levels.

A GymAware (Kinetic Performance Technologies, Canberra, Australia) linear position transducer (LPT), was used to monitor velocity of all repetitions. Two LPTs were placed on either side of the barbell, perpendicular to the position between the hands and the loaded barbell sleeves, per the manufacturer's instructions. The reliability and validity of this LPT was previously confirmed and its characteristics described elsewhere [22,23]. The GymAware v2.8.0 app was used on a tablet (iPad, Apple Inc., California, USA) to transmit data from the LPT via Bluetooth. The LPT attached to the right side of the barbell was connected to a TV providing visual feedback of mean barbell velocity for each repetition. Thus, all analyzes are based on this LPT's data. To avoid issues with cloud data storage or internet connectivity, mean velocity of all repetitions was manually recorded and organised in an excel spreadsheet (Microsoft Corporation, Redmond, Washington, USA). Additionally, to ensure consistency and accuracy, this was always completed by the same two researchers.

To assess personality traits, the 50-item International Personality Item Pool (IPIP) Big Five Personality Inventory was used [24,25]. This inventory contains 50 questions, 10 for each personality dimension (Agreeableness, Conscientiousness, Extraversion, Neuroticism, and Openness). IPIP items were administered via 5-point, Likert-type scales ranging from 1 (very inaccurate) to 5 (very accurate). Scores were obtained by averaging the scores for each item. Only conscientiousness and emotional stability were analysed.

2.7. Statistical analysis

Linear regression models – with maximal velocity of each set as a predictor and XRM as the outcome – were used to determine individual and general XRM-velocity relationships. General relationships were determined separately for each testing session by pooling data from all participants, whereas individual relationships were obtained for each participant separately, on both testing sessions. The goodness of fit of general relationships was examined though the coefficient of determination (R^2) and residual standard error (*RSE*) of the models, whereas medians and ranges of R^2 and *RSE* were evaluated for individual XRM-velocity relationships.

Multiple linear regression models were used to examine influential factors on individual XRM-velocity relationships' goodness of fit. Load (3 levels), training experience (2 levels) and practices (2 or 3 levels), relative strength and conscientiousness and emotional stability were all considered predictors.

The predictive validity of the general and individual XRM-velocity relationships was examined by using the models from the first testing session and fitting them to the second testing session data. The general and individual models' errors were then evaluated by calculating an absolute difference between the observed and predicted data in the second testing session. Models were deemed practically useful for monitoring and prescribing RT if the error rates were below 2 repetitions. To examine factors affecting the probability of exceeding a prediction error of more than 2 repetitions, multiple logistic regression models were used, specified with a binomial distribution and a logit link function.

For linear models, predictors' estimates and 95% confidence intervals (95%CI) were calculated and presented whereas, for generalised models, odds ratios with associated 95%CI were evaluated and presented to aid interpretation. For categorical predictors with more than 2 levels, post-hoc Holm-Bonferroni corrections were performed. Additional models' specifications and diagnostics details are in Supplementary File II.

All statistical analyzes were performed in R language and environment for statistical computing (version 4.2.0, The R Foundation for Statistical Computing, Vienna, Austria) using the *stats* package, models' performance using the *performance* package [26], and preparation and visualisation of data using the *tidyverse* [27] and *sjPlot* [28] packages. Custom-written R script and associated dataset are available at the Open Science Framework repository (URL: https://osf.io/pqa3v/).

3. Results

Goodness of fit for the general XRM-velocity relationship was strong and comparable on the first ($R^2 = 0.45$; $RSE = 3.44$ repetitions; Fig. 1) and second ($R^2 = 0.49$; $RSE = 3.66$ repetitions; Fig. 1) testing sessions. However, the goodness of fit of the individual XRM-velocity relationships was always stronger on both the first ($R^2 = 0.98$ [0.58, 1.00]; $RSE = 1.01$ [0.07, 5.5] repetitions; Fig. 1) and second ($R^2 = 0.98$ [0.50, 1.00]; $RSE = 1.15$ [0.12, 4.24] repetitions; Fig. 1) testing sessions. R^2 of the individual XRM-velocity relationships was not affected by any of the examined factors ($p \geq 0.295$; Table 1). Individual models' RSE was only affected by sex, with males having lower *RSE* than females ($p = 0.033$; Table 1).

The predictive validity of the general XRM-velocity relationship was not acceptable since the absolute error between the observed and predicted data on the second testing session – using the model established in the first testing session – was higher than 2 repetitions on average (mean

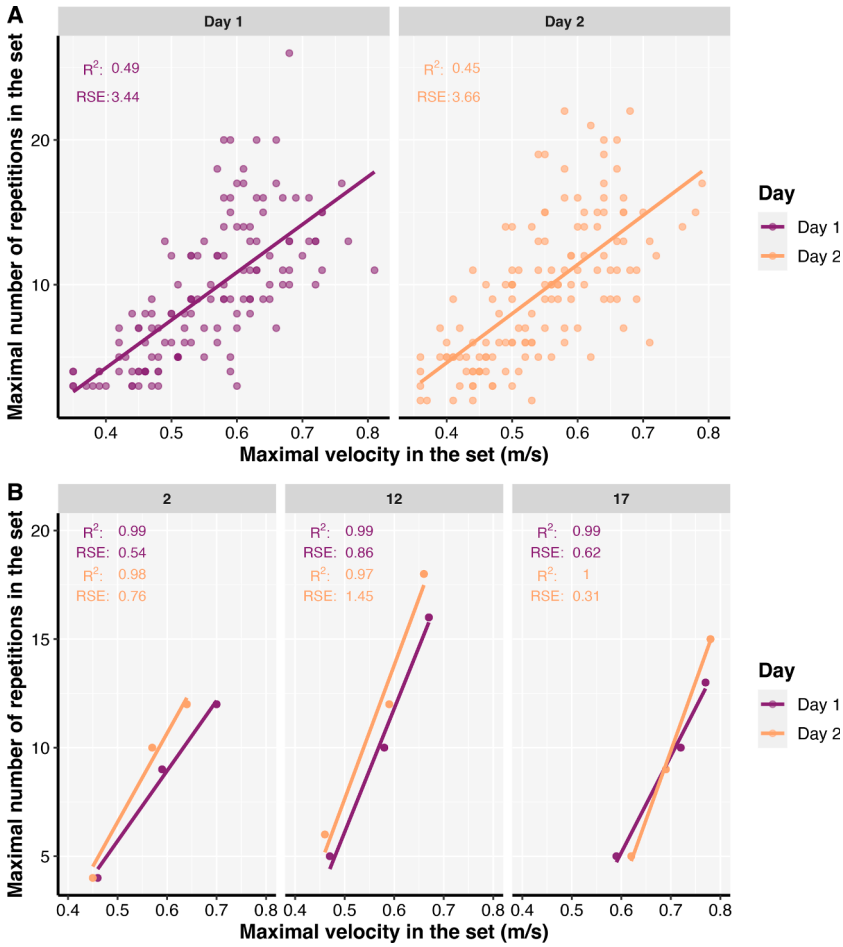

Fig. 1. General (A) and individual (B) relationships between the absolute velocity of the fastest repetition in a set and the maximum number of repetitions completed in a set. Coefficient of determination (R^2), as well as residual standard errors (RSE), are also presented for general and individual relationships established in both testing sessions. Note that data for the individual relationships are from three representative participants.

Table 1
Factors affecting the goodness of fit of individual XRM-velocity relationships.

	R[2]			RSE		
Predictors	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*
(Intercept)	0.99	0.79–1.19	< 0.001	1.27	-0.83–3.36	0.232
Day [Day 2]	-0.00	-0.04–0.04	0.984	-0.07	-0.47–0.34	0.739
Sex [male]	0.03	-0.03–0.09	0.389	-0.70	-1.34–-0.06	0.033
Emotional Stability	-0.00	-0.00–0.00	0.483	0.02	-0.01–0.06	0.225
Conscientiousness	-0.00	-0.01–0.00	0.295	-0.00	-0.04–0.03	0.866
Training experience [> 3 years]	0.02	-0.02–0.07	0.328	0.10	-0.40–0.60	0.686
Loads practices [70 – 80% 1RM]	0.03	-0.03–0.09	0.325	0.19	-0.44–0.82	0.553
Loads practices [> 80% 1RM]	0.04	-0.03–0.11	0.271	0.32	-0.40–1.04	0.380
Repetitions practices [8–12]	0.01	-0.04–0.06	0.734	-0.13	-0.69–0.43	0.648
Repetitions practices [> 12]	-0.01	-0.07–0.05	0.727	0.62	-0.00–1.24	0.051
Repetitions in reserve practices [> 2 RIR]	0.02	-0.03–0.06	0.476	0.22	-0.28–0.72	0.383
Relative strength (1RM/BM)	-0.01	-0.07 – 0.06	0.844	-0.27	-0.98 – 0.43	0.445
Observations	92			92		
R^2 / R^2 adjusted	0.131 / 0.012			0.220 / 0.112		

Note: Reference groups were the following: Load [70% 1RM], Sex [female], Training experience [< 3 years], Loads practices [< 70% 1RM], Repetitions practices [< 8 repetitions], Repetitions in reserve practices [< 2 RIR]. 1RM, one repetition maximum; BM, body mass; RIR, repetitions in reserve; OR, odds ratio; R^2, coefficient of determination; RSE, residual standard error; CI, 95% confidence intervals; p, *p* value; XRM-velocity relationship, relationships between the number of repetitions completed to failure and the maximum velocity in the set.

ε = 2.81 repetitions; Supplementary file IV). Individual XRM-velocity relationships displayed acceptable predictive validity as the absolute error of 2 repetitions, although being close to it, was not exceeded on average (mean ε = 1.96 repetitions; Supplementary File IV). Multiple linear regression models did not reveal any influential factors affecting absolute errors of the individual XRM-velocity relationships in

predicting data on the second testing session. In addition, multiple logistic regression models did not reveal any influential factors affecting the probability of exceeding an absolute prediction error of 2 repetitions for individual XRM-velocity relationships (Table 2). A more detailed descriptive analysis of prediction errors broken down by sex, training experience, and relative strength is presented in Supplementary File IV.

4. Discussion

This is the first examination of the goodness of fit and prediction accuracy of general and individual XRM-velocity relationships during free-weight back squats while exploring the effects of sex, training status and history, as well as personality traits on goodness of fit and accuracy of those relationships. The main findings were (1) individual XRM-velocity relationships provided a higher goodness of fit compared to general relationships on both testing sessions; (2) the goodness of fit of individual XRM-velocity relationships was not affected by sex, training status and history or personality traits; (3) individual, but not general XRM-velocity relationships established on the first testing session provided acceptable predictions of XRM in the second testing session, on average; and (4) prediction accuracy of individual XRM-velocity relationships was not affected by sex, training status and history, or personality traits. Therefore, these findings expand on previous studies on XRM-velocity relationships in Smith machine exercises and support the utility of individual XRM-velocity relationships with free-weight exercises in practice, regardless of the sex, training status and history, or personality traits of trainees.

Velocity-based RT monitoring and prescription has many applications. The most researched applications are the use of (1) load-velocity profiles which allow for training load prescription that accounts for individuals' daily readiness to train, and (2) the concept of velocity loss (VL) to manage fatigue and set-volume with stipulations to perform repetitions until a pre-determined VL is exceeded [29]. Indeed, a strong relationship exists between VL magnitude and the percentage of completed repetitions out of the maximum possible [30,31]. Thus, researchers recommended each training set should be terminated when a given VL is reached instead of prescribing a given, fixed, number of repetitions per set [30]. However, authors of more recent studies outlined several methodological issues with using VL in practice [32,33]. Specifically, these authors noted high inter-individual variability in the number of completed repetitions before reaching different VL thresholds, suggesting the inability of VL thresholds to control for set-volume across individuals with adequate prevision [32,33]. To address this issue, Garcia-Ramos et al. [16] and Miras-Moreno et al. [17] proposed the use of maximal velocities across sets to failure to predict XRM and control for set-volume in Smith machine exercises. In line with their findings [16,17], the goodness of fit for individual XRM-velocity relationships was considerably greater in the present study, with R^2 values twice as high for the individual compared to general XRM-velocity relationships in the free-weight back squat. Moreover, the current study also investigated factors affecting the goodness of fit of individual XRM-velocity relationships. Only sex affected the goodness of fit, with males' XRM-velocity relationships having a slightly lower *RSE* ($p = 0.033$) compared to females. However, the median *RSE* for females was still low and comparable to males (*RSE* = 1.42 vs 0.94), with almost identical R^2 values for the model's fit ($R^2 = 0.97$ vs 0.98). Considering this, and the lack of significant effects from training status, history, and personality traits on the stability of individual XRM-velocity relationships, these findings could likely be generalised to the resistance-trained population using free-weight exercises. Importantly, participants in the present study performed only 3 sets to failure, compared to the previous studies [16,17] where participants performed 4 sets to failure to establish the XRM-velocity relationship. Thus, the testing procedure for the XRM-velocity relationship could be simplified by performing only 3 sets to failure with 90, 80, and 70% of 1RM, as in the present study. Since the XRM-velocity relationship seems to be linear, it is likely that this relationship can be further simplified by examining only two loads (i.e., the two-point method). Though, this is yet to be investigated.

Although high goodness of fit is important for judging the usefulness of any model, it does not guarantee its predictive validity. Researchers previously reported high goodness of fit of XRM-velocity relationships on two separate days and concluded it could be used to reliably, and with high precision, estimate different XRMs in Smith machine exercises [16]. However, such findings do not necessarily confirm the predictive validity of XRM-velocity relationships since they were established using the data from the respective sessions. Modelling XRM-velocity relationships would be useful for RT monitoring and prescription purposes, and more beneficial compared to no cost, traditional RT prescription methods, only if they can predict future XRM data reasonably well. Therefore, the present study investigated the ability of general and individual XRM-velocity relationships established on the first testing session to predict XRM in the subsequent testing session. We observed general XRM-velocity relationships had unacceptable prediction errors of 2.81 repetitions, on average. In contrast, individual XRM-velocity relationships demonstrated lower, acceptable (i.e., < 2 repetitions) prediction errors, on average. Neither sex, training status,

Table 2
Factors affecting the (1) the absolute differences between predicted and observed XRM in a subsequent testing session based on individual XRM-velocity relationships; and (2) the probability of individual XRM-velocity relationships exceeding a prediction error of 2 repetitions.

	Absolute differences			Exceeding the error		
Predictors	*Estimates*	*CI*	*p*	*OR*	*CI*	*p*
(Intercept)	1.40	-1.47–4.27	0.336	1.15	0.03–56.13	0.942
Sex [male]	0.10	-0.78–0.98	0.823	0.76	0.23–2.47	0.649
Relative strength (1RM/BM)	-0.35	-1.32–0.63	0.482	0.52	0.13–1.92	0.330
Emotional Stability	0.04	-0.01–0.09	0.092	1.04	0.98–1.11	0.228
Conscientiousness	0.00	-0.05–0.05	0.976	0.98	0.91–1.05	0.484
Training experience [> 3 years]	0.03	-0.66–0.71	0.935	1.14	0.46–2.93	0.774
Repetitions practices [8 – 12]	-0.23	-1.00–0.55	0.565	0.83	0.28–2.45	0.736
Repetitions practices [> 12]	0.78	-0.07–1.64	0.072	2.68	0.87–8.58	0.088
Loads practices [70 – 80% 1RM]	-0.46	-1.33–0.41	0.297	0.78	0.25–2.35	0.652
Loads practices [> 80% 1RM]	-0.72	-1.71–0.27	0.151	0.56	0.15–2.01	0.378
Repetitions in reserve practices [> 2 RIR]	-0.00	-0.69–0.68	0.990	1.42	0.55–3.73	0.465
Observations	138			Observations		138
R^2 / R^2 adjusted	0.162 / 0.096			R^2 Tjur		0.121

Note: Reference groups were the following: Load [70% 1RM], Sex [female], Training experience [< 3 years], Loads practices [< 70% 1RM], Repetitions practices [< 8 repetitions], Repetitions in reserve practices [< 2 RIR]. 1RM, one repetition maximum; BM, body mass; RIR, repetitions in reserve; OR, odds ratio; R^2, coefficient of determination; R^2 Tjur, Tjur's pseudo coefficient of determination for generalised linear models with binary outcomes; RSE, residual standard error; CI, 95% confidence intervals; p, p value; XRM-velocity relationship, relationships between the number of repetitions completed to failure and the maximum velocity in the set.

history, or personality traits affected the accuracy of individual XRM-velocity relationships. Considering all the above, individual, but not general XRM-velocity relationships demonstrate acceptable accuracy and may be useful for RT monitoring and prescription.

Although the prediction errors of individual XRM-velocity relationships were acceptable, on average, caution should be exercised when implementing this concept in practice as models' prediction errors for some individuals might still be unacceptable. For instance, Fig. 2 illustrates 8 individuals' data. For half of them, models' prediction errors were on average almost perfect (< 1 repetition; Fig. 2A), but for the other half, prediction errors exceeded as many as 4 repetitions (Fig. 2B). Looking more closely at those individuals did not reveal any unique or explanatory factors differentiating them from one another. Namely, they had similar training practices and personality traits, and in both groups, there were 2 females and males. The only noticeable, albeit minor, difference was observed in relative strength, with the group of 4 individuals who had lower prediction errors being slightly stronger (relative strength = 1.56vs 1.40 1RM/body mass). However, since multiple regression models did not reveal any effects of strength on XRM-velocity relationship prediction accuracy, it is more likely that relative strength does not affect prediction accuracy, rather some other unexplored factor(s) may. Interestingly, when the mean prediction accuracy of XRM-velocity relationships was broken down by sex, strength levels and training experience, models for less experienced individuals consistently yielded higher errors than for more experienced individuals (Supplementary File IV). Additionally, this was exaggerated among less strong individuals, especially males. Therefore, practitioners should evaluate XRM-velocity relationships on a case-by-case basis and determine their usefulness, especially among less experienced individuals, to safeguard against potential, unwanted prediction errors which can lead to erroneous training decisions and suboptimal training adaptations.

In the present study, the goodness of fit and predictive validity of XRM-velocity relationships for monitoring and prescribing RT were examined while also exploring the effects of a range of factors on these relationships. Based on the findings reported in this study, practitioners can predict XRM daily without frequent RTF testing using only the fastest repetition in a set which allows them to prescribe RT loads (e.g., 8RM), and at the same time, to have an insight into the training volume completed and the number of repetitions left in reserve after the set is completed. However, there are several considerations when interpreting the results. First, whether these findings transfer to other free-weight exercises is unknown. However, the effects of training history and status, sex, and personality traits are likely applicable to a variety of exercises. Second, all participants had ≥6 months of RT experience which may explain why training experience did not affect the stability or accuracy of the XRM-velocity relationships. However, it is unclear whether these findings can be extrapolated to those without RT experience. Third, there were fewer females in this study than males. However, the females had a wide range of strength levels, training experience, and training practices, improving the sample's generalisability. Finally, notably, the acceptable prediction error choice for the threshold of XRM-velocity relationships may have influenced our findings and interpretation thereof. However, any method with an error greater than 2 repetitions may not be justifiable in practical settings, given other well-established, cost-free methods of monitoring and prescribing RT set-volume exist, such as stopping sets at a predetermined perceived number of repetitions in reserve.

5. Conclusion

The present study expands on previous XRM-velocity relationship studies while providing several novel findings, specifically for free-weight exercises. Individualised XRM-velocity relationships provided twice as high goodness of fit and were able to predict XRM in a subsequent testing session with acceptable accuracy compared to general XRM-velocity relationships. Although sex, training status and history, and personality traits did not affect the goodness of fit of individual XRM-velocity relationships or their prediction accuracy, some individual XRM-velocity relationships might still yield unacceptable prediction errors. Therefore, XRM-velocity relationship accuracy should be evaluated on a case-by-case basis and future studies should explore additional factors which might cause prediction errors in some individuals. Nevertheless, an individual XRM-velocity relationship can be established with a simple linear regression model using the maximal velocity

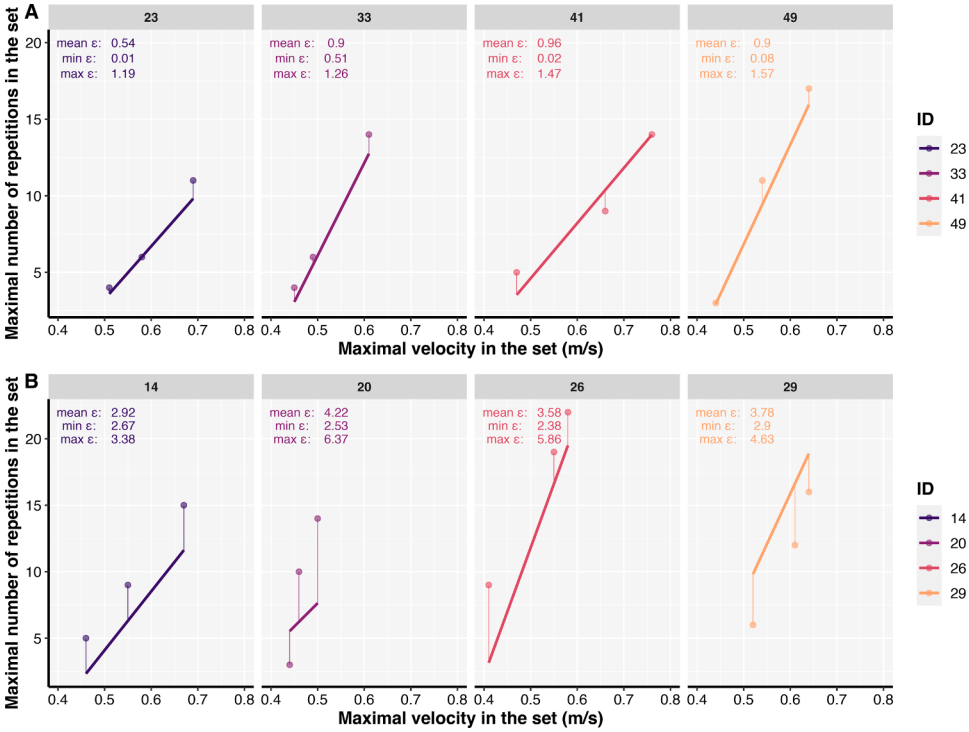

Fig. 2. Individual relationships between the absolute velocity of the fastest repetition in a set and the maximum number of repetitions completed in a set. Thin vertical lines connecting thick lines and dots across participants represent distances (or residual errors) between predicted and observed data. Mean (mean ε), minimal (min ε), and maximal (max ε) errors are also presented for each participant and load. This data is from representative participants where models' prediction errors never (A) and always (B) exceeded (A) a 2 repetitions error mark.

of three sets to failure with 90, 80 and 70% of 1RM and the corresponding number of repetitions performed to failure. This allows practitioners to prescribe RT loads matching an intended XRM by recording only the fastest velocity of the set (usually within the first 3 repetitions) with a given load and predicting, with reasonable accuracy, an individual's XRM.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.physbeh.2023.114158.

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